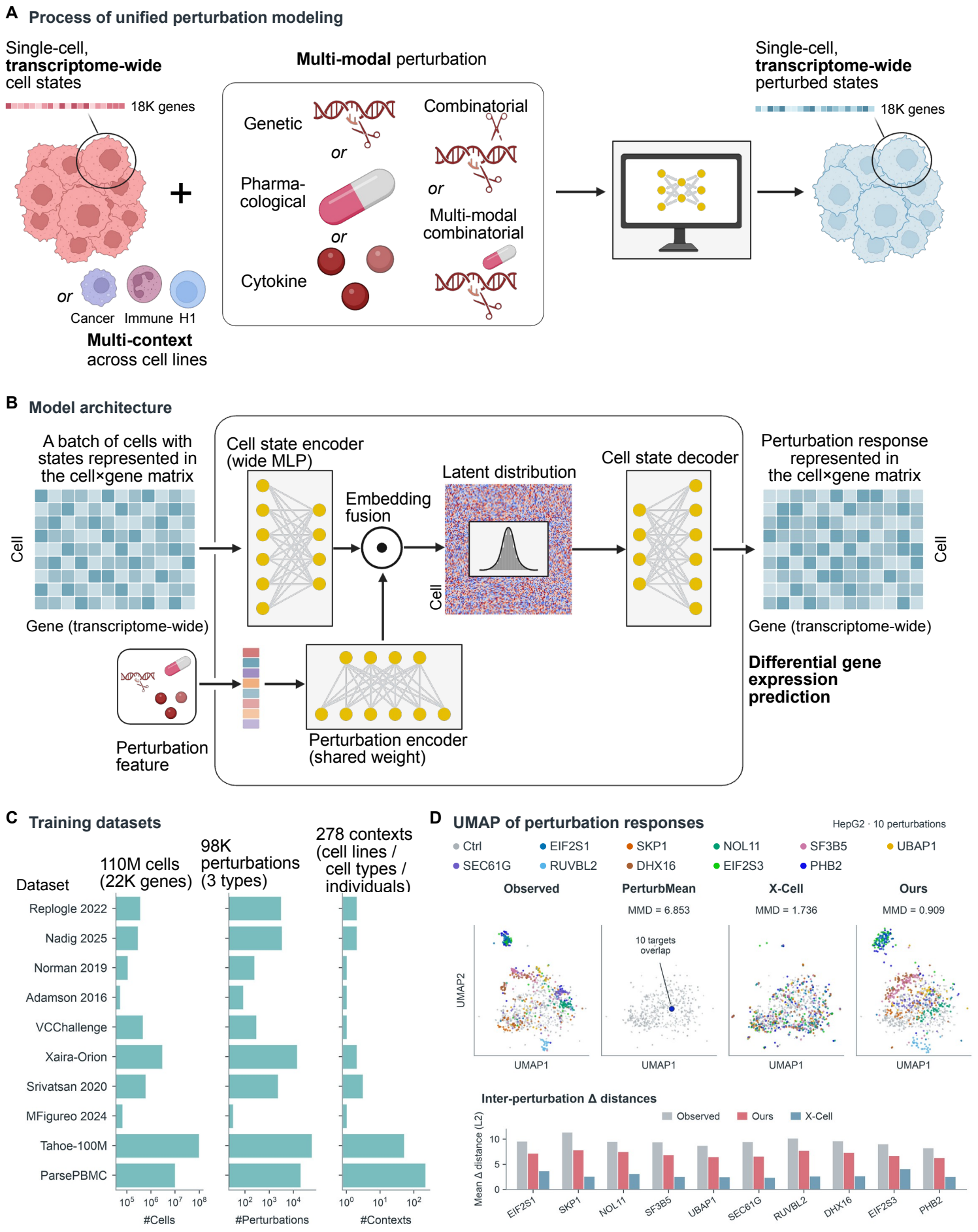

A
Process of unified perturbation modeling
Single-cell,
transcriptome-wide
cell states
18K genes
or
Cancer
Immune
H1
Multi-context
across cell lines
Multi-modal perturbation
Genetic
or
Pharma-
cological
or
Cytokine
Combinatorial
or
Multi-modal
combinatorial
Single-cell,
transcriptome-wide
perturbed states
18K genes
B
Model architecture
A batch of cells with
states represented in
the cell×gene matrix
Cell
Gene (transcriptome-wide)
Cell state encoder
(wide MLP)
Embedding
fusion
Latent distribution
Cell
Cell state decoder
Perturbation response
represented in
the cell×gene matrix
Cell
Gene (transcriptome-wide)
Differential gene
expression
prediction
Perturbation
feature
Perturbation encoder
(shared weight)
C
Training datasets
Dataset
110M cells
(22K genes)
98K
perturbations
(3 types)
278 contexts
(cell lines /
cell types /
individuals)
Replogle 2022
Nadig 2025
Norman 2019
Adamson 2016
VCChallenge
Xaira-Orion
Srivatsan 2020
MFigureo 2024
Tahoe-100M
ParsePBMC
#Cells
#Perturbations
#Contexts
D
UMAP of perturbation responses
HepG2 · 10 perturbations
Ctrl
EIF2S1
SKP1
NOL11
SF3B5
UBAP1
SEC61G
RUVBL2
DHX16
EIF2S3
PHB2
Observed
PerturbMean
MMD = 6.853
X-Cell
MMD = 1.736
Ours
MMD = 0.909
10 targets
overlap
UMAP2
UMAP1
Inter-perturbation Δ distances
Observed
Ours
X-Cell
Mean Δ distance (L2)
EIF2S1
SKP1
NOL11
SF3B5
UBAP1
SEC61G
RUVBL2
DHX16
EIF2S3
PHB2